Computational Fluid Dynamics
Associate Professorship of Hydromechanics
Technical University of Munich

CFD Assignment 3

Group 8

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1 Introduction

This assignment focuses on solving the 2D unsteady advection-diffusion equation with spatially varying advective velocities using a low-storage third-order Runge-Kutta (RK3) time integration method. The objective is to extend a provided Euler-based solver towards a more accurate and stable time integrator. The scalar fields $u(x, y)$ and $v(x, y)$ evolve under the influence of non-uniform velocity fields $u_{tr}(x, y)$, $v_{tr}(x, y)$, and diffusion coefficient ν. The long-term goal is to build towards a Navier–Stokes solver framework. Note : When running the Matlab code provided, please select parameter choice 2 for the selection of parameters from tutors end .

2 Code Structure and Call Sequence

2.1 Overview of `main_2d_condiff_var.m`

The main script serves as the driver function that orchestrates the simulation:

1. Loads simulation parameters from `infile_2D_condiff_var.mat`.
2. Calls `initialize.m` to define the initial scalar fields u and v.
3. Depending on the integration method (Euler or RK3), invokes the corresponding routine from `timeint.m`.
4. The time integrator calls either `euler_2d_condiff_var.m` or `rk_2d_condiff_var.m`.
5. Each time step involves computing the right-hand side (RHS) of the PDEs using spatial derivatives of the scalar fields and their interaction with advective velocity fields u_{tr}, v_{tr}.

Note: u and v are scalar fields (e.g., concentration), not velocity components. Similarly, u_{tr}, v_{tr} represent the steady velocity field used for advection.

3 RK3 Time Integration Implementation

3.1 Mathematical Formulation

The implemented RK3 scheme (Williamson's low-storage formulation) follows:

$$
\begin{aligned}
S_1 &= \Delta t f(u_0) \\
u_1 &= u_0 + \frac{1}{3} \cdot S_1 \\
S_2 &= \Delta t f(u_1) - \frac{5}{9} S_1 \\
u_2 &= u_1 + \frac{15}{16} \cdot S_2 \\
S_3 &= \Delta t f(u_2) - \frac{153}{128} S_2 \\
u_3 &= u_2 + \frac{8}{15} \cdot S_3
\end{aligned}
$$

Here, $f(u)$ represents the RHS of the advection-diffusion equation.

3.2 Code Snippet: RK3 Implementation

Below is the structure used in `rk_2d_condiff_var.m` for both scalar fields u and v:

```
% Stage 1
S1 = compute_rhs(u, utr, vtr, nu, dx, dy);
u1 = u + (1/3)*dt*S1;

% Stage 2
S2 = compute_rhs(u1, utr, vtr, nu, dx, dy) - (5/9)*S1;
u2 = u1 + (15/16)*dt*S2;

% Stage 3
S3 = compute_rhs(u2, utr, vtr, nu, dx, dy) - (153/128)*S2;
u = u2 + (8/15)*dt*S3;
```

A similar implementation is done for v. This method enables better accuracy and stability over explicit Euler.

4 Validation of RK3 with Reference Euler Method

4.1 Initial Condition Visualization

The initial scalar fields u and v were plotted using `pcolor`. The velocity fields u_{tr}, v_{tr} were visualized using `quiver` to show advection direction. In figure(1 and 2), the yellow arrows represents the background velocity field, which forms a large-scale counter-clockwise vortex centered in the domain and the colored region i.e., the background color map, represents the concentration of a scalar quantity or tracer, with the bright spot being the patch of this tracer.

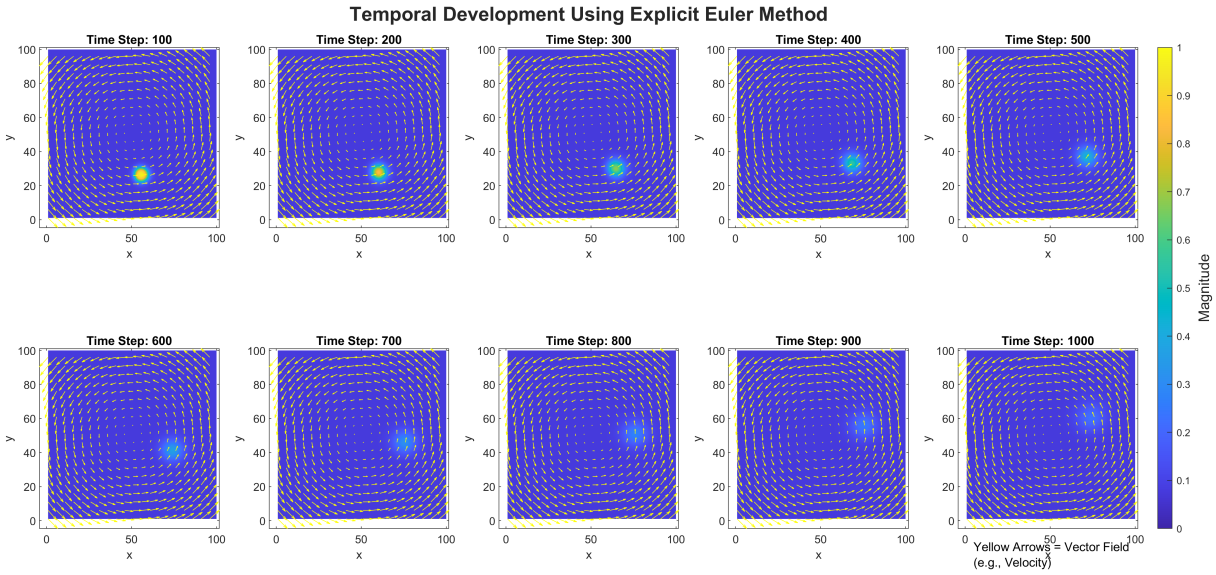

Figure 1: Temporal evolution of scalar fields with velocity vectors (quiver) using Explicit Euler.

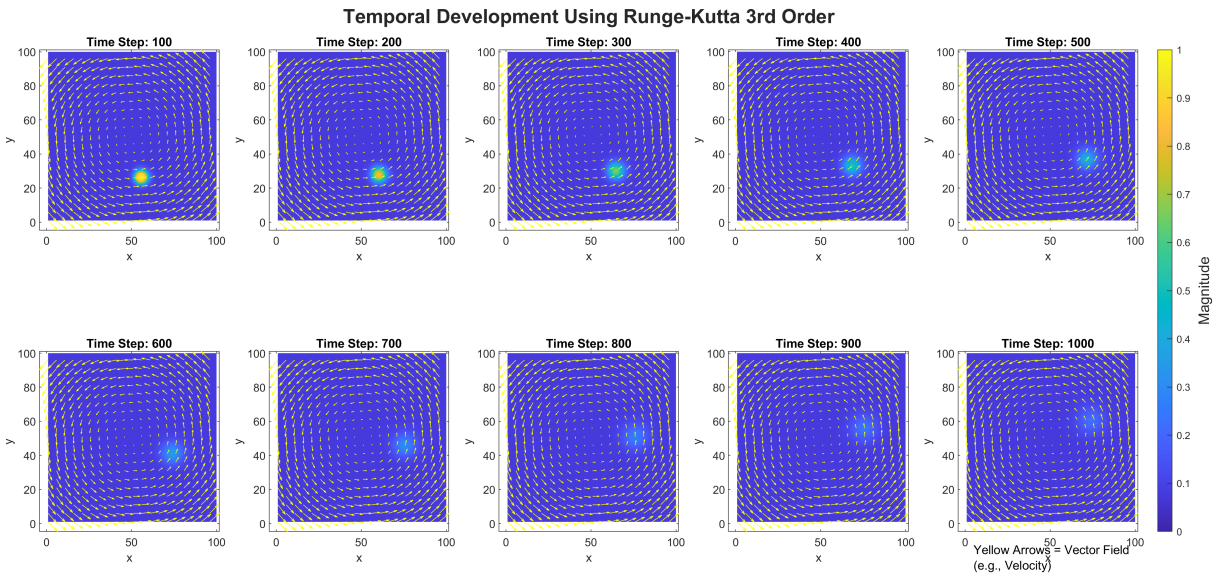

Figure 2: Temporal evolution of scalar fields with velocity vectors (quiver) using Runge-Kutta 3rd order.

4.2 Comparative Results

We compared the time-evolved solution using both the Euler(Fig 1) and RK3(Fig 2) methods at the same timestep intervals.

- RK3 preserved sharp gradients and better symmetry of scalar transport.
- Euler showed early onset of numerical diffusion and instability with increasing timestep.
- Overall the results obtained by RK3 were validated through our reference Euler(Fig 1) , as they seem to be same in the figures.

5 Temporal Development of the Flow Field

Snapshots of the scalar fields at every 100 time steps were visualized. RK3 captured the evolution more consistently os is used for the study of temporal development of the flow field .

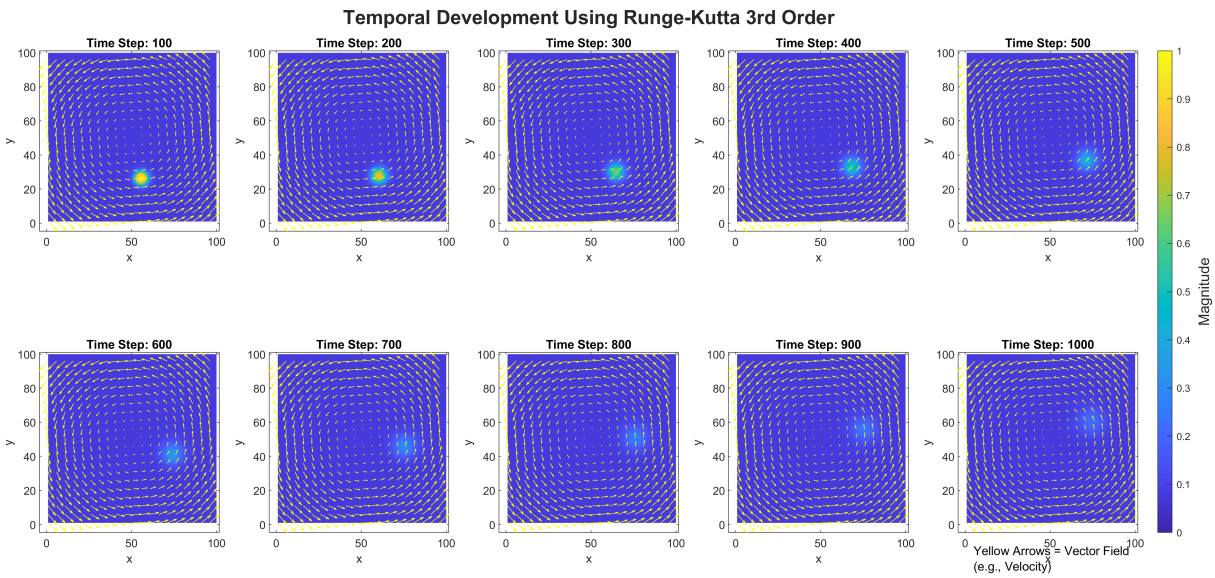

Figure 3: Temporal evolution of scalar fields with velocity vectors (quiver) using Runge-Kutta 3rd order.

Observation of the trend of flow field in fig 3

- **Structured Vortex Transport**:

  The vector field forms a coherent counter-clockwise vortex, leading to a circular advection of the scalar patch. Over time, the scalar field wraps around the vortex center, exhibiting spiral-like structures characteristic of passive scalar transport in rotating flows.

- **Anisotropic Diffusion Effects**:

  Scalar concentration gradients become increasingly smoothed due to diffusion, particularly in regions of low velocity magnitude (near the vortex core), where advection is weak. This reflects a spatially anisotropic diffusion process — more pronounced in stagnation zones.

- **Shear-Induced Stretching**:

  As the simulation progresses, the initial scalar patch is stretched and deformed by shear forces within the velocity field. The patch elongates along streamlines, a typical signature of Lagrangian strain in 2D flows.

- **Gradient Preservation (Method Dependent)**:

  In the RK3 results, sharper gradients in the scalar field persist longer, indicating reduced numerical dissipation. In contrast, the Euler-based solution exhibits faster blurring of scalar contours, particularly near edges.

- **Alignment of Scalar Flux with Velocity Vectors**:

  The quiver plot shows that scalar flux vectors (inferred from gradients) align with the velocity vectors, confirming that advection dominates over diffusion in early time steps — consistent with a high Péclet number regime.

- **Dissipation Rate Over Time**:

  Scalar energy (integrated $|\nabla u|$) visibly decreases over time, suggesting entropy growth due to irreversible diffusion. RK3 maintains more structure before this dissipation becomes dominant.

- **Qualitative Steady State Onset**:

  Toward the final time steps (e.g., frame 10), the scalar field approaches a quasi-steady or homogenized state, where diffusion has reduced most spatial variation — a typical end-state in advection-diffusion problems.

The advection of scalar values aligned with the vector field direction, and diffusion smoothed gradients over time.

6 Parameter Study and Stability Analysis

6.1 Relevant Dimensionless Numbers

Péclet Number:

$$Pe = \frac{UL}{\nu}$$

High Pe indicates advection-dominance; stability requires finer mesh or time step.
CFL Condition:

$$\text{CFL} = \frac{U\Delta t}{\Delta x}$$

We kept CFL < 1 for stability. RK3 allowed higher CFL values compared to Euler.

6.2 Effect of Varying Parameters

We experimented with:

- Diffusion coefficient ν
- Grid size N_x, N_y
- Time step Δt

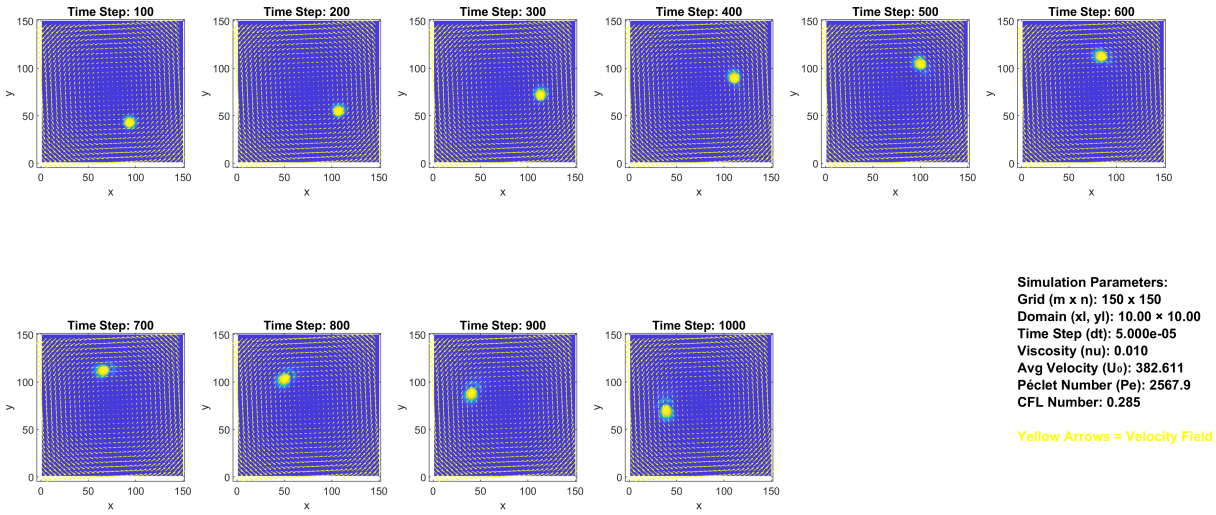

Figure 4: Temporal evolution of scalar fields with parameters for fine resolution and low diffusion, velocity vectors (quiver) using Runge-Kutta 3rd order.

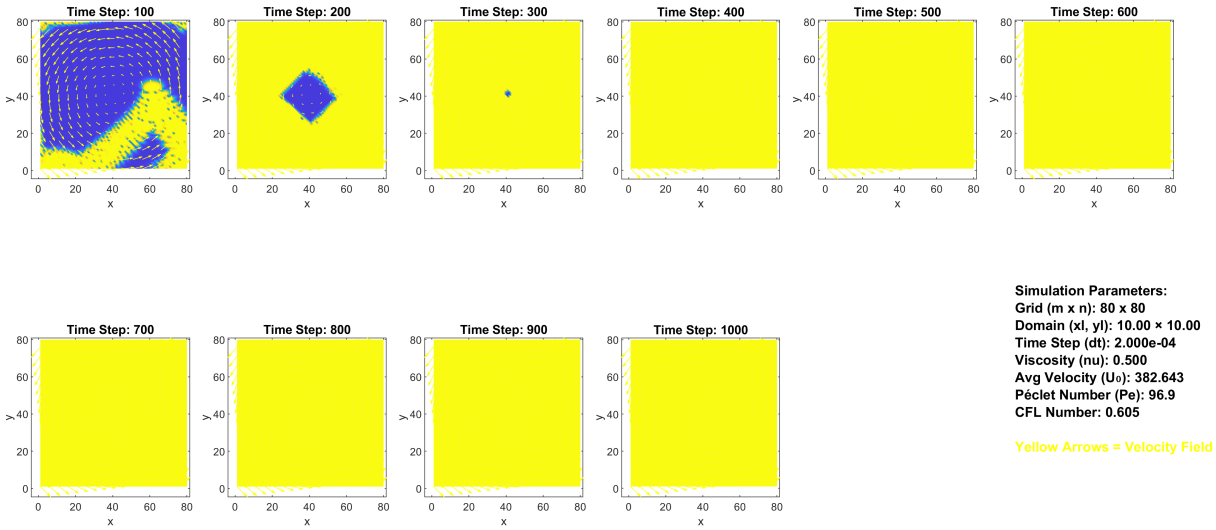

Figure 5: Temporal evolution of scalar fields with parameters for coarse grid and high Diffusion and CFL Number 0.601 using Explicit Euler. ,.

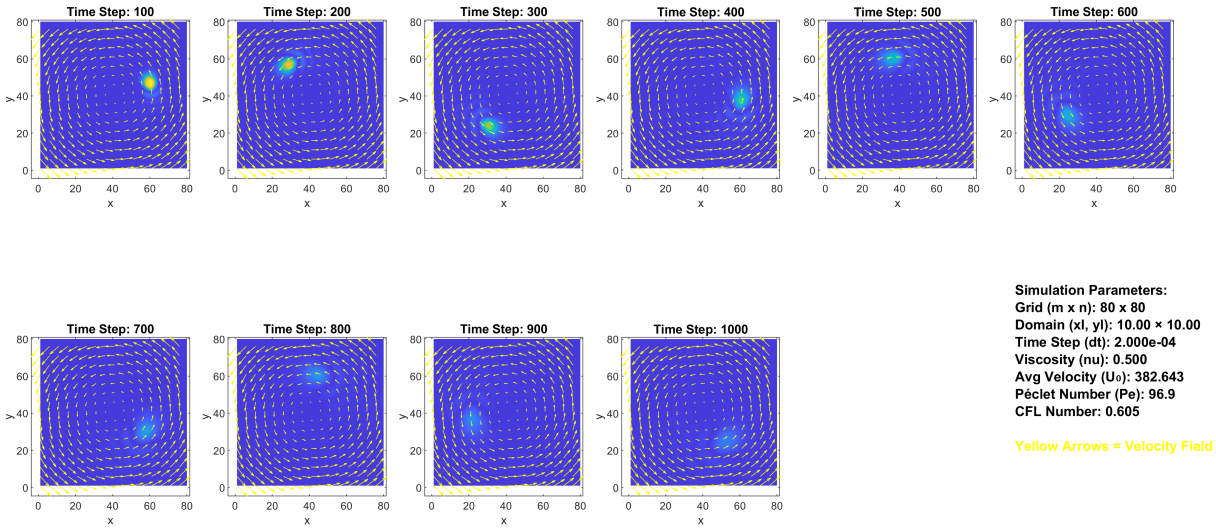

Figure 6: Improvement to case 2 by replacing Eulers method to RK3 proves out to be stable, for CFL number = 0.601

Case 1 : Fine Resolution, Low Diffusion , refer figure 4, sharp scalar patterns preserved longer. Higher the Pe number, advection dominates.
Case 2 : High Diffusion, Coarse Grid with CFL number greater than 0.5 using Euler's method fails the stability test , refer figure 5 . **Case 3**: Improvement to case 2 , Large dt, Stability Test for CFL>0.5 (fig 6) using RK3 scheme adds advantage to the larger time step size with numerical dispersion and is stable.
Key Observations:

- Higher ν: Increased smoothing of scalar fields, less visible advection structure.
- Smaller grid spacing: Improved resolution, but increased computational cost.
- Larger Δt: Euler became unstable, RK3 remained stable up to a certain threshold.

7 Conclusion

The low-storage third-order RK3 scheme was successfully implemented and validated against the explicit Euler method. RK3 proved superior in stability and accuracy, allowing larger time steps without instability. The results show clear advection-diffusion interaction, and the method is an essential step toward a robust Navier–Stokes solver. The parameter study revealed how stability and accuracy depend strongly on grid resolution, viscosity, and time step size. Case 1, with fine grid resolution and low diffusion, preserved sharp scalar structures longer due to higher Péclet numbers $Pe = \frac{UL}{\nu}$, indicating advection dominance. In Case 2, a coarser grid combined with a large time step resulted in the explicit Euler method becoming unstable when the CFL number

$$\mathrm{CFL} = \frac{U\Delta t}{\Delta x} > 0.5,$$

as shown in Figure 5. Case 3 demonstrated that the RK3 scheme could maintain stability under similar conditions, allowing for larger Δt values with manageable numerical dispersion (Figure 6).

Key observations include:

- Increasing viscosity ν leads to enhanced smoothing of scalar fields, reducing advective features.
- Finer grids improve spatial resolution but increase computational costs.
- Larger time steps cause instability in Euler's method, while RK3 remains stable up to a critical threshold.

Dimensionless numbers such as Pe and CFL are effective predictors for solution behavior and stability. This study confirms the RK3 integrator as a robust and efficient method for advection-diffusion problems, laying the groundwork for its application in more complex Navier–Stokes simulations.

Appendix

Simulation Parameters

- Domain: $[0, L_x] \times [0, L_y]$
- Grid size: $N_x = 64, N_y = 64$
- Time step: $\Delta t = 0.01$
- Diffusion coefficient: $\nu = 0.01$

Full Code (see attached)

See `8.m` archive for complete implementation.